



MGMT90280 Managerial Decision Analytics

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Table of Contents

1. Introduction	3
1.1 Organizational Background.....	3
1.2 Description of Business Problem.....	3
2. Literature Review	4
2.1 Synopsis	4
2.2 Strengths	4
2.3 Weaknesses	6
2.4 Summary.....	6
3. Methodology	7
4. Implementation.....	9
4.1 Descriptive Analysis.....	9
4.2 Predictive Analysis	13
4.2.1 Cancellation Rate Forecast	13
4.2.2 Passenger Carried Forecast.....	16
4.3 Prescriptive Analysis.....	20
4.3.1 Integer Linear Programming	20
4.3.2 Integer Linear Programming Model.....	20
4.3.3 Model Construction and Assumptions	21
4.3.4 Decision Variables and Objective Function	23
5. Discussion	25
5.1 Strengths of Analysis	25
5.1.2 Leveraging Real Company and Industry’s Data.....	25
5.1.3 Alignment to Practice.....	26
5.2 Limitations of Analysis	26
5.3 Recommendations	27
5.3.1 Dynamic Adjustment of Flight Schedules	27
5.3.2 Strengthen Operational Resilience and Cost Control	28
6. Conclusion.....	28
References.....	29
Appendices.....	32

1. Introduction

1.1 Organizational Background

Qantas was founded in 1920 in the Queensland outback and has grown over the years to be Australia's largest domestic and international airline (Qantas, 2025b). With the symbol of the "Flying Kangaroo", Qantas enjoys over 65% of the domestic market share, and currently possesses a fleet of over 300 aircraft and has about 23,000 non-executive staff, with a domestic market share of over 65% (QantasNewsRoom, 2025). Data from 2024 shows that the three main routes (Melbourne-Sydney, Gold Coast-Sydney, Perth-Sydney) contribute 45% of the total passenger volume for the top 10 domestic routes which account for 68% of total passenger volume. The Melbourne-Sydney route offers over 9.2 million seats each year, which is about 25,000 passengers per day (Woodley, 2025). The performance of these three routes varies quite a bit. The Melbourne-Sydney route has a cancellation rate of 12% due to high passenger demand, while the Perth-Sydney and Gold Coast-Sydney routes have cancellation rates of 7% and 4%, respectively (BITRE, 2025a). These routes are essential for Qantas' domestic revenue and highlight the imbalance between supply and demand, making them the main focus of this analysis.

1.2 Description of Business Problem

As Australia's leading airline with a large scale, Qantas Airways is currently facing operational and reputational challenges owing to imbalance between demand and supply, high flight cancellation rates, declining On-Time Performance and intense arrangements of labour. In 2023-2024, the operational problems caused by "inaccurate demand prediction & mismatch in flight numbers" at Qantas resulted in significant losses - during this period, they paid out 120 million Australian dollars in customer compensation. In the Qantas Group's 2024 fiscal year report, it was reported that the quantity of voluntary employee turnover rate was 6.5%, further weakening the stability of the capacity (Qantas, 2025a) This problem has shown differentiated outbreaks on the three core routes: Melbourne-Sydney, Gold Coast-Sydney, and Perth-Sydney under the focus of this study: the high-demand routes experienced increased delays due to "insufficient capacity", while the medium and low-demand routes suffered from "excessive flights" resulting in resource waste. Both factors jointly pushed up operational costs.

As the tourism season from July to December 2025 approaches (with the passenger flow expected to increase by 15% compared to the previous year during the Christmas holiday in December) (7NEWS, 2024). If there is still a lack of precise predictions of passenger volume and cancellation rates for the three routes, and if flight numbers are planned only based on experience, it may repeat the problem of "peak season capacity gap & off-peak empty flights waste", thus the purpose of the report is utilizing predictive analysis to clarify demand and risks, and then formulate the optimal flight allocation plan for the second half of 2025 through normative analysis to achieve the goal of "cost reduction - ensuring punctuality - maintaining capacity stability".

Various sources of numerous data including Qantas Annual Report, ACCC Monitoring Report, World Bank, and BITRE OTP Status are collected to support the research. The plan for the report begins with a literature review, which discusses the abstract, strengths and weaknesses of selected articles, then approaches of methodology and implementation, with an analysis including descriptive, predictive and prescriptive analyses and concludes with suggestions for improvement and evaluation of Qantas' analytical approach.

2. Literature Review

2.1 Synopsis

We started with a literature review, researching the use of analytical methods in the airline industry, and how these methods can help Qantas to improve the challenges it is facing. The literature was mainly a review of peer-reviewed journal articles published in the last 10 years to date, drawn from databases such as ScienceDirect, Google Scholar, ResearchGate, and university libraries. The key words used were "ILP Models," "Clustering," and "Time Series." These steps ensured the academic robustness and methodological soundness of the literature we have investigated.

2.2 Strengths

This group of papers provides different methods, usage and approaches, and gives important information for our analysis and research. For example, Lonzius and Lange (2024) use clustering models to classify. It is possible to deduce similarity and difference in patterns of

operation of different types of aircraft. The findings of the cluster analysis will assist in our great improvement in conceptions of the types of operations carried on by different types of aircraft, and routes and conditions of operation. Furthermore, in the use of information such as passenger loads and cancellation rates during the classification process it will not only assist in giving information of the differences between types of aircraft, but give better and more useful information for investigations.

K-means model described by Yu et al. (2022) has several advantages to apply to our data on Australian airlines. It is illustrated that the model might be applied when analysing aircraft data of the various airlines, clustering the data according to passenger behaviour and flight operation data. Thus, airlines may be able to classify customers so that certain marketing may be undertaken to distinguish parts of clientele and success rate of service recommendations improved. Similarly, also Lee et al. (2019) portrays the question of modeling under uncertainty, as it applies the Regression Model in conjunction with the evaluation of a point merge system (PMS) holding algorithm. This study has illustrated how the algorithm helps to reduce delays and eliminate manual work, giving information how the influence of the Qantas delays may be controlled.

Abdelghany et al. (2025) has applied Time series analysis, using models of SARIMA, VAR and SARIMAX in forecasting passenger demand for airline routes. They have also reviewed the effect of seat capacity and fare structure on the accuracy of prediction of the various algorithms used. The main advantage of applying this approach to Australian airline study is that it allows the company to analyse patterns of capacity demand expressed in terms of historical flight data patterns.

Zhou et al. (2020) applied an Integer Linear Programming (ILP) model, a basic procedure for solving a greater part of the principal problems concerned with matters applicable to airline operations, such as fleet assignment planning (FAP), aircraft routing planning (ARP), and crew scheduling (CSP). By the establishing of the decision variables in the form of integer variables and making use of linear constraints the model operated to maximize the possible airline operations. This is a method that allows us to sufficiently match with phenomenal accuracy the capacity to be obtained by different types of aircraft operated point to point by the Qantas lines and demand on routes obtainable, and at a minimum expense of aircraft-route allocation.

Finally, Ahmed et al. (2018) modeled the RIARCP problem, which becomes a Mixed Integer Linear Programming (MIIP) model solvable with the aid of solvers. When applied to KLM on

a large scale, Air France, and some other airlines' data they successfully derived from the model solutions of maximum sorts, with the result of improving interference reductions, the positive practical importance in robust scheduling for complex airline processes.

2.3 Weaknesses

While these articles give very useful ideas, models and analytical tools, they also highlight some gaps in how they can be applied to our target concentration on the Qantas three domestic routes and cutting operational costs. For example, Zhou et al. (2020) and Ahmed et al. (2018) are very theoretical and do not include practical examples, so these papers are not seen as particularly useful for the unique situation of Qantas. Also, the works like Lonzius et al. (2024) and Yu et al. (2022) look more at abnormal operations or grouping of passengers, rather than the fleet assignment model or cost reduction models that we need. Also, some of the models like Lee et al. (2019) are based on assumptions like continuous demand or simple scheduling structures. These may not fit the realities of the actual domestic market conditions in Australia. All in all, it seems that while these studies give good indicator of appropriate optimization methods these need to be adjusted to consider the uniqueness of Qantas high frequency domestic routes particularly about real demand patterns and the competitive market in the domestic air transport sector.

2.4 Summary

All the literature reviews helped us to conduct deep analysis on Qantas in various perspectives, including k-means which used to segment passenger groups and identify travel patterns, regression models supported demand estimation and highlighted key variables influencing booking behaviour. Moreover, integer linear programming was applied to optimize flight scheduling under operational constraints. By applying these models contained in the reviews, our analysis will be more comprehensive and insightful. Results through applying descriptive, predictive and prescriptive analysis will be shown in the methodology part.

3. Methodology

This section elaborates the adoption of a 3-stage analysis framework: Descriptive, Predictive, Prescriptive Analysis to solve Qantas' mismatch between operating and demand. The aim of this structure is to systematically diagnose problems, forecast future's operating key metrics, and generate final executive optimal flight allocation plan. The main data sources for this research are mainly from public data of the Australian Infrastructure, Transport and Regional Economy Bureau (BITRE), Qantas' annual report and other public market data. This three-stage approach ensures our recommendations are data-driven and actionable. The complete analytical pathway is summarised in Table 1 below.

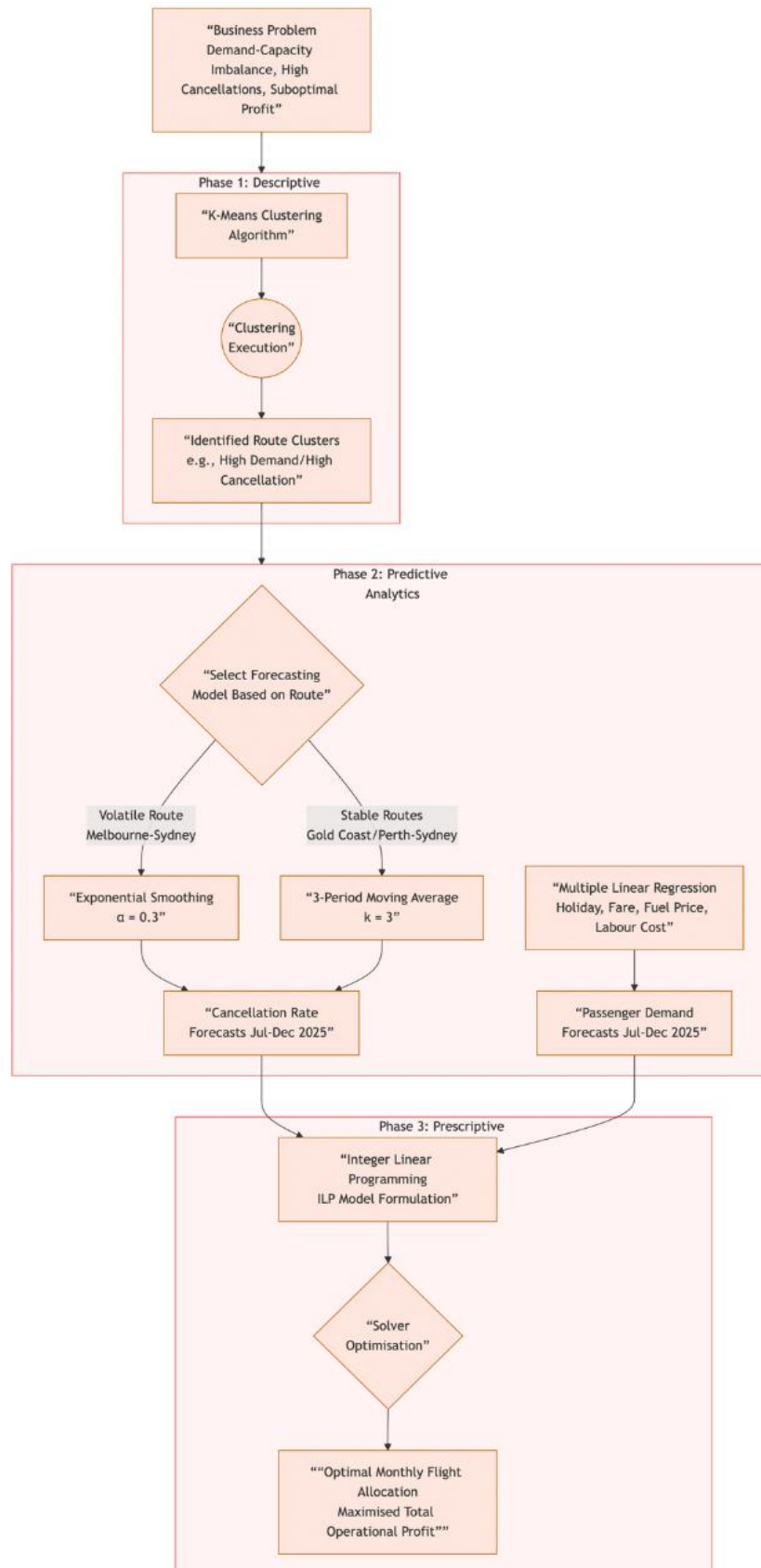


Table 1. Analytical Workflow for Qantas Flight Optimisation

The process of the analytics is depicted in the above flowchart, which shows the analytic processes that were undertaken, beginning with descriptive analytics, in which a k-means clustering analysis was undertaken to cluster Qantas's major domestic routes based on passenger volume and cancellation rates. The second stage, predictive analytics, involved a combination of analytic techniques, including time series forecasting and multiple linear regression in order to arrive at accurate forecasts for future passenger demand and cancellation rates for the second half of 2025. Reasonable assumptions were made regarding future operational conditions. The last stage, prescriptive analytics, resulted in the tactics suggested in this paper, such that the ideal means of distributing flights among routes and months in order to maximise total operational profit could be estimated by employing an Integer Linear Programming model solved with OpenSolver.

4. Implementation

4.1 Descriptive Analysis

Clustering has been widely applied in airline transport research because it is helpful to capture seasonal demand trends and operational performance in large datasets. Yu et al. (2022) demonstrated the efficiency of k-means to group passengers' demand and actionable insights provided for route allocation in the airline industry. Lonzius and Lange (2024) also employed clustering to classify aircraft routings and evaluate their relationship with passenger carried and flight on time performance. Thus, k-means clustering analysis was adopted to classify Qantas' major domestic routes, aiming to analyze different traffic volumes and cancellation rates across time.

In order to minimize the effects of lockdown and policy constraint during COVID-19 on air traffic control, the selected dataset covers 30 months (Jan 2023 - Jun 2025) operations on the domestic routes. Instead of examining all top domestic routes, three representative routes were selected based on geographical and operational diversity across Australia (Melbourne-Sydney, Perth-Sydney, Gold Coast-Sydney). Passenger volume and cancellation rates were chosen as key variables as they reflect the demand intensity and service performance. Due to data privacy constraints, Qantas's specific passenger carried were not directly disclosed by the airline. Instead, industry-level monthly data was obtained from the Bureau of Infrastructure and

Transport Research Economics (BITRE). An approximate number of Qantas' passengers carried was derived accordingly:

Passengers carried (Qantas approx.) per month of each route

$$= \text{Passengers carried (industry wide)} \cdot \frac{\text{Qantas actual number of flights}}{\text{All Airlines number of flights}}$$

This approach provides a reliable demand data estimation of Qantas, thereby facilitating cluster analysis.

Figure 1, 2, and 3 presents the monthly cancellation rate for each route from January 2023 (1) to December 2024 (24). Apparently, the Melbourne - Sydney route contains the highest and most volatile cancellation rate, which is fluctuating between approximately 4% to 12%. However, there is no consistent upward or downward trend observed across the three routes, suggesting that cancellation rates might be influenced by many external factors rather than seasonality effect. These observations suggest that future demand forecasting should consider multiple explanatory variables rather than time only, which is consistent with findings of Dayioglu and Alnipak (2023).

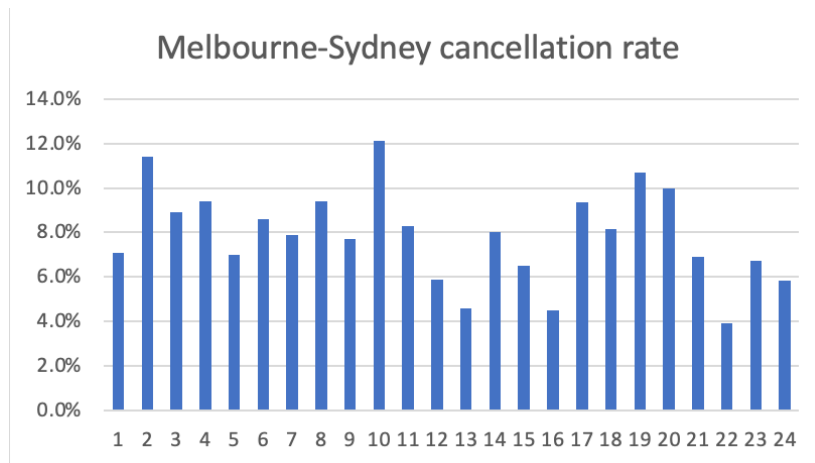


Figure 1. Cancellation Rate of Melbourne—Sydney

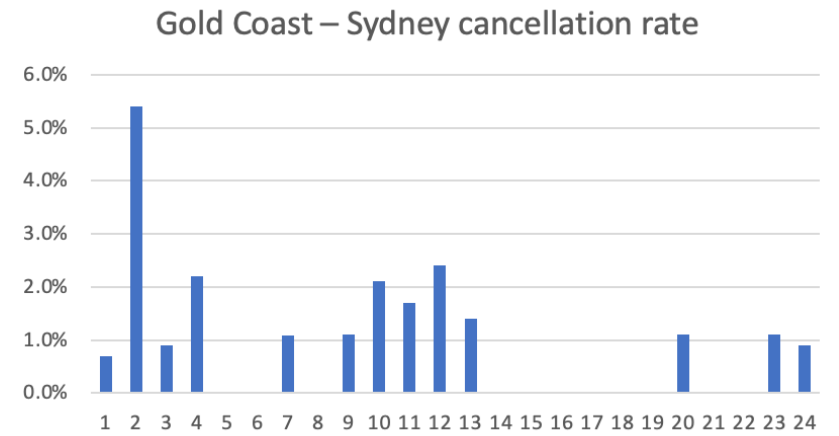


Figure 2. Cancellation Rate of Gold Coast—Sydney

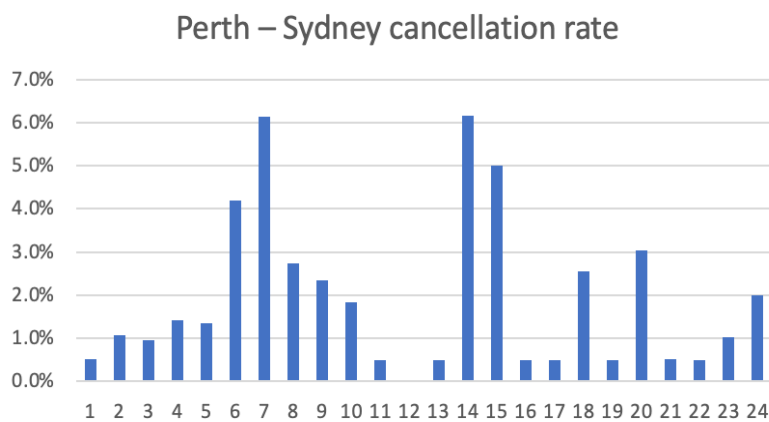


Figure 3. Cancellation Rate of Perth—Sydney

The data of passenger volume and cancellation rate of each route from January 2024 to December 2024 were used in clustering. Passenger volumes were rescaled in order to better distinguish demand difference across routes, while cancellation rates were standardized through z-score to ensure clustering results reflect the relative fluctuation rather than absolute scale. This step allows the clustering analysis to reveal meaningful patterns of demand scale and operational reliability among Australia’s major domestic routes. As shown in Figure 4, the clustering was performed using k-means to obtain three clusters and ten iterations were adopted.

Variables		
# Selected Variables	2	
Selected Variables	rescaled passenger	z_cancellation

Rescaling: Fitting Parameters	
Rescale Data?	FALSE

K-Means Clustering: Fitting Parameters	
# Clusters	3
# Iterations (Max)	10
# Starts	5
Random Seed	12345

K-Means Clustering: Reporting Parameters	
Show Cluster Centers	TRUE
Show Random Starts Summary	TRUE
Show Inter-Cluster Distances	TRUE
Show Clusters Summary	TRUE
Show Fitting Metrics	TRUE
Show Record-Cluster Distances	FALSE

K-Means Clustering: Scoring Parameters	
Score Training Partition	TRUE

Figure 4. K-means Algorithm

The clustering results are shown below in Figure 5. The first cluster represents routes with low traffic and stable operations (Rescaled passenger: 1.5; z_cancellation: -0.81532), characterized by small passenger volumes and below average cancellation rates which mainly correspond to Gold Coast–Sydney. The cluster 2 mainly comprising the Perth–Sydney route, represents medium demand routes with moderate cancellation risk (rescaled passenger: 3.1667; z_cancellation: -0.3636), where operational challenges might be caused by seasonal factors (Guyen et al., 2019). The third cluster includes a high demand route with high cancellation rate (rescaled passenger: 13.333; z_cancellation: 1.179), which suggests large passenger volumes such as Melbourne–Sydney route may experience more disruptions due to capacity constraints or scheduling challenges.

Cluster Centers

Cluster	rescaled passenger	z_cancellation
Cluster 1	1.5	-0.815322345
Cluster 2	3.166666667	-0.36369803
Cluster 3	13.33333333	1.179020375

Figure 5. Cluster Centers

The clustering results provide the basis for the following analysis by grouping monthly route operations into different demand and cancellation categories. This section is useful to understanding how the intensity of demand and the volatility of cancellations jointly shape the operational dynamics and develop differing benchmarks for optimizing flight allocations. These benchmarks offer guidance for prescriptive models, in assigning greater numbers of flights to demand-intensive and volatile routes, while constraining frequencies of stable but low-demand routes.

4.2 Predictive Analysis

4.2.1 Cancellation Rate Forecast

After confirming significant differences in demand volatility and operational stability across the three routes, our next objective is to forecast passenger carried and flight cancellation rates for the period from July to December 2025. From seasonal variability, alternative tests were carried out for all periods, knowing that no statistically significant seasonal changes happen during this 30-month observation. Hence, no seasonal adjustment was made. Additionally, selecting the type of model mainly relies on the data's characteristics. For example, the Melbourne-Sydney (MS) route has a moderate upward trend with short-term fluctuations, so exponential smoothing ($\alpha=0.3$) is an effective way to balance responsiveness and noise interference. On the other hand, the Gold Coast-Sydney (GS) and Perth-Sydney (PS) routes have stable random fluctuations, thus a three-period moving average ($k=3$) is more suitable. In terms of α and k , we used the iterative algorithm to search for the optimal path and minimize the Mean Squared Error (MSE) of validation data.

Time	Melbourne-Sydney	Gold Coast – Sydney	Perth – Sydney
1	7.10%	0.70%	0.51%
2	11.40%	5.40%	1.08%
3	8.90%	0.90%	0.95%
4	9.40%	2.20%	1.42%
5	7.00%	0.00%	1.35%
6	8.60%	0.00%	4.19%
7	7.90%	1.08%	6.13%
8	9.40%	0.00%	2.74%
9	7.70%	1.11%	2.34%
10	12.10%	2.10%	1.84%
11	8.30%	1.70%	0.51%
12	5.90%	2.40%	0.00%
13	4.60%	1.40%	0.49%
14	8.00%	0.00%	6.15%
15	6.50%	0.00%	5.00%
16	4.48%	0.00%	0.51%
17	9.33%	0.00%	0.49%
18	8.14%	0.00%	2.56%
19	10.71%	0.00%	0.49%
20	10.00%	1.10%	3.03%
21	6.90%	0.00%	0.51%
22	3.90%	0.00%	0.50%
23	6.74%	1.10%	1.03%
24	5.82%	0.90%	1.99%
25	5.60%	0.00%	1.93%
26	5.60%	0.00%	2.13%
27	3.00%	10.50%	0.50%
28	5.80%	1.60%	0.51%
29	5.60%	0.00%	0.00%
30	6.90%	0.00%	0.51%

Table 2. Historical Cancellation Rate for 2023 Jan-2025 Jun

First, the pronounced volatility of MS route time series is commensurate with its slight non-stationarity: when $\alpha = 0.3$, the forecast obtained by the model for the second half of 2025 is between 5.87% and 8.87%. That is, it can be averaged to 6.62%. A lower result was achieved from the historical mean of 7.38% (see Tables 2 and 3). This suggests a gradual recovery in route activities and enhanced assurance in flight schedules during post-pandemic capacity reallocation. However, there might be a bias in long-term forecasts in the absence of such seasonal adjustments. Additionally, the absence of considering possible seasonal peaks (e.g., December) may affect the accuracy of short-term projections (Alaoui et al., 2025).

Time	Melbourne-Sydney
July	5.87%
August	5.87%
September	7.37%
October	5.87%
November	5.87%
December	8.87%
Average	6.62%

Table 3. Melbourne-Sydney Cancellation Rate Forecast for 2025 Jul-Dec

Second, both GS and PS routes operate on a three-period moving average model ($k=3$). This approach is used to predict the cancellation rate of the next period by averaging the most recent three observations. As shown in Table 4, the model yielded average forecast values of 0.30% and 0.33%, effectively smoothing out random fluctuations and generating relatively stable forecast results. The sharp decline from historical averages (1.14% and 1.71%) likely reflects both smoothing tendency and increased reliability. However, due to the lower complexity, the model would experience a short-run lag bias and might underestimate the effect of abrupt changes in operations or external shocks.

Time	Gold Coast – Sydney	Perth – Sydney
July	0.53%	0.34%
August	0.18%	0.28%
September	0.24%	0.38%
October	0.32%	0.33%
November	0.25%	0.33%
December	0.27%	0.35%
Average	0.30%	0.33%

Table 4. Gold Coast-Sydney & Perth-Sydney Cancellation Rate Forecast for 2025 Jul-Dec

Overall, the forecasting framework is methodologically consistent and provides empirically valid short-term insights. For Qantas, the distinctive modeling approach could offer data-driven insights and improves the explanatory value of cancellation patterns. However, long-term reliability may be limited by the lack of seasonal adjustment and external variables (such as

weather, maintenance, or capacity shifts). Therefore, future research could adopt more flexible hybrid models, such as Holt–Winters or ARIMA, to improve the accuracy and adaptability of predictions (Ram et al., 2025).

4.2.2 Passenger Carried Forecast

Rank	Route	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25
1	Melbourne-Sydney	280,021	300,063	291,196	296,341	326,906	269,550
2	Gold Coast-Sydney	32,839	31,474	29,906	31,986	39,945	51,604
3	Perth-Sydney	48,238	48,719	51,614	52,597	54,383	54,322

Table 5. Passenger Carried Forecast of Three Routes for 2025 Jul-Dec

The second section focuses on the forecast of Qantas passenger carried. Table 5 shows the forecast results for the three routes based on the time variable. The results show that the Melbourne-Sydney route will maintain Qantas' busiest route in the second half of 2025. Meanwhile, demand for Perth-Sydney will grow steadily, while demand for Gold Coast-Sydney will likely show an important seasonal uptrend. In addition, Figures 6, 7 and 8 show that the p-values for the three routes are all less than 0.05. It shows that the passenger carried forecast results based on the time variable are robust.

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.667572316							
R Square	0.445652797							
Adjusted R Square	0.425854683							
Standard Error	11855.70842							
Observations	30							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	1	3163936998	3163936998	22.5098607	5.57702E-05			
Residual	28	3935619022	140557822.2					
Total	29	7099556020						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	247800.9243	4439.640442	55.81553902	2.97142E-30	238706.7331	256895.1155	238706.7331	256895.1155
X Variable 1	1186.489948	250.0792508	4.744455785	5.57702E-05	674.2258248	1698.754072	674.2258248	1698.754072

Figure 6. Melbourne-Sydney Time Variable

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.76							
R Square	0.578							
Adjusted R Square	0.563							
Standard Error	657.3							
Observations	30							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	1	9E+07	9E+07	3.331118254	0.078666595			
Residual	28	7E+08	3E+07					
Total	29	8E+08						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	29284	1928	15.19	4.79E-15	25334.65108	33234.27303	25334.65108	33234.27303
X Variable 1	198.2	108.6	1.825	1.69E-06	-24.25069396	420.7248764	-24.25069396	420.7248764

Figure 7. Gold Coast-Sydney Time Variable

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.671273105							
R Square	0.450607582							
Adjusted R Square	0.430986424							
Standard Error	2143.415026							
Observations	30							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	1	105508246.6	105508246.6	22.96539205	4.89458E-05			
Residual	28	128638383.3	4594227.975					
Total	29	234146629.9						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	42914.58409	802.6506468	53.46608049	9.79971E-30	41270.42877	44558.7394	41270.42877	44558.7394
X Variable 1	216.6672907	45.21228128	4.792222037	4.89458E-05	124.0541308	309.2804505	124.0541308	309.2804505

Figure 8. Perth-Sydney Time Variable

However, to provide more accurate decision-making and optimization strategies for prescriptive analysis. In addition to considering the single time variable mentioned above, we also applied a multiple regression model to evaluate and explain the influence of various variables on Qantas' passenger demand. According to Srisaeng (2015) and Zachariah et al. (2023), the primary determinants of Australia's domestic LCCs demand are real best discount airfare, population, real GDP, real GDP per capita, labour cost, world jet fuel prices, holiday, and tourism attractiveness. Through data collection and analysis, we found that some variables lacked data, so we selected the following variables. Specifically, the independent variables include Holiday, Fare, Fuel Price, and Labor Cost. The dependent variable is Passenger Carried. The model is as follows:

$$Passenger_t = \beta_0 + \beta_1 Holiday_t + \beta_2 Fare_t + \beta_3 Fuel Price_t + \beta_4 Labour Cost_t$$

Specifically, the Holiday variable is used to account for seasonal holiday effects. Through the collection and analysis of historical data, we identified January, April and December as holiday months, assigning them a value of 1, while the remaining months were assigned a value of 0. Fare represents ticket prices (unit AUD). Additionally, Labor Cost is expressed in millions of Australian dollars.

However, based on the actual operational results for the three routes (see Figures 9, 10, and 11), we first observe that the R^2 value for the Melbourne-Sydney route is approximately 52%. The values for the Gold Coast-Sydney and Perth-Sydney routes are approximately 33% and 22%, respectively. This indicates that the model has stronger explanatory power for the Melbourne-Sydney route than for the other two routes. Furthermore, the p-values for the three routes reveal that the fare variable for both the Melbourne-Sydney and Gold Coast-Sydney routes is < 0.05 . In other words, passenger volume on these two routes is significantly affected by fare fluctuations. Furthermore, the holiday variable for the Melbourne-Sydney route is also < 0.05 (0.00026), showing that holiday months have an important influence on passenger volume on this route (Maneejuk & Yamaka, 2021). However, this model has limitations, passenger volume may be affected by a variety of factors (Ethan et al., 2025). Furthermore, the sample size, covering only two and a half years of data. It is relatively limited and importantly weakening the model's explanatory power.

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.718872317							
R Square	0.516777408							
Adjusted R Square	0.43624031							
Standard Error	16630.79012							
Observations	29							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	4	7098936591	1774734148	6.416638016	0.001167292			
Residual	24	6637996321	276583180					
Total	28	13736932912						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	100905.8155	99382.30728	1.015329773	0.320075749	-104209.1856	306020.8165	-104209.1856	306020.8165
Holiday	-32314.16749	7564.465487	-4.271837521	0.000264569	-47926.45693	-16701.87805	-47926.45693	-16701.87805
Fuel Price	-10416.45548	7003.710937	-1.487276613	0.149963056	-24871.40441	4038.493448	-24871.40441	4038.493448
Fare	1024.41844	468.5636616	2.186295105	0.038780064	57.35057269	1991.486307	57.35057269	1991.486307
Labour Cost	147.4068753	95.76596187	1.539240795	0.136827378	-50.2443557	345.0581062	-50.2443557	345.0581062

Figure 9. Melbourne-Sydney Passenger Carried Multiple Regression

SUMMARY OUTPUT								
<i>Regression Statistics</i>								
Multiple R	0.517795787							
R Square	0.268112477							
Adjusted R Square	0.146131224							
Standard Error	5489.632197							
Observations	29							
<i>ANOVA</i>								
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>			
Regression	4	264953962.9	66238490.74	2.197980993	0.09962578			
Residual	24	723265479.9	30136061.66					
Total	28	988219442.8						
	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
Intercept	-15417.10543	46631.79551	-0.330613592	0.743805351	-111660.4011	80826.19025	-111660.4011	80826.19025
Holiday	3046.614605	4316.500824	0.705806562	0.487103634	-5862.205238	11955.43445	-5862.205238	11955.43445
Fuel Price	-249.5743871	2287.428646	-0.109106961	0.914024792	-4970.595079	4471.446304	-4970.595079	4471.446304
Fare	289.8815309	292.9222707	0.989619294	0.332233218	-314.6803223	894.4433841	-314.6803223	894.4433841
Labour Cost	13.98561713	30.4447791	0.459376535	0.650097693	-48.84931866	76.82055293	-48.84931866	76.82055293

Figure 10. Gold Coast-Sydney Passenger Carried Multiple Regression

SUMMARY OUTPUT								
<i>Regression Statistics</i>								
Multiple R	0.472202073							
R Square	0.222974797							
Adjusted R Square	0.093470597							
Standard Error	4149.616403							
Observations	29							
<i>ANOVA</i>								
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>			
Regression	4	118589931.5	29647482.87	1.721757261	0.178057813			
Residual	24	413263591.1	17219316.3					
Total	28	531853522.6						
	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
Intercept	34916.66311	13818.32637	2.526837344	0.018512032	6397.03919	63436.28703	6397.03919	63436.28703
Holiday	509.857583	1977.912073	0.257775656	0.798777543	-3572.352299	4592.067465	-3572.352299	4592.067465
Fuel Price	2403.606569	1725.225869	1.393212687	0.176326248	-1157.084621	5964.297759	-1157.084621	5964.297759
Fare	60.04256193	30.26988958	1.983573867	0.058856269	-2.431419641	122.5165435	-2.431419641	122.5165435
Labour Cost	21.02725658	23.55029839	0.892865824	0.380794797	-27.57817038	69.63268354	-27.57817038	69.63268354

Figure 11. Perth-Sydney Passenger Carried Multiple Regression

4.3 Prescriptive Analysis

4.3.1 Integer Linear Programming

Looking back at the Descriptive analysis, we identified that the key challenges in the company's operations lie in: how to efficiently allocate fixed flight number under highly fluctuating demand, in order to avoid two types of waste: the waste of empty capacity when demand is insufficient, and the waste of lost opportunities when demand is high.

In the "Forecasting" section as shown in Table 6, we have been able to predict the monthly demand (number of passengers) and operational risks (cancellation rate) for each route from July to December in 2025. However, these predictions alone do not directly create value. The core task of this section is to convert these predictive results into actionable business decisions: Determine the optimal number of monthly flights on each major domestic route based on the forecasted passenger demand, so as to meet the market demand while maximizing the company's total profit.

4.3.2 Integer Linear Programming Model

Combining the results derived from both current and future demand forecasts of Qantas's key routes, this then led us to our final stage of optimisation. Thanks to the demand and cancellation rate forecasts, it was necessary that we were able to determine for each route and for each month the ideal frequency with which the company should operate its services in order to meet the forecast demand, consistent with both profitability and the financial and operational limitations of the company.

A capacity analysis was made (as shown in Table 6) in order to obtain a better idea of the existing company resources to meet the estimated demand. The analysis indicated a need for better management of Qantas machinery resources, which may include the flexible allocation of aircraft or working out of scheduled frequencies. Therefore, a prescriptive analysis should also provide useful indications with a view to enhancing the effectiveness of the company.

Route	Month	Predicted Demand	Capacity to demand
Mel-Syd	Jul	280021	80%
	Aug	300063	80%
	Sep	291196	75%
	Oct	296341	75%
	Nov	326906	70%
	Dec	269550	70%
Gol-Syd	Jul	32839	80%
	Aug	31474	80%
	Sep	29906	75%
	Oct	31986	75%
	Nov	39945	70%
	Dec	51604	65%
Per-Syd	Jul	48238	95%
	Aug	48719	95%
	Sep	51614	95%
	Oct	52597	95%
	Nov	54383	95%
	Dec	54322	95%

Table 6. Capacity Analysis (2025 Jul-Dec Forecast vs. Historical Capacity)

4.3.3 Model Construction and Assumptions

1. Demand Fulfilment Constraint

The demand fulfilment requirements were set that the percentage of forecasted demand fulfilled by each route-month (as shown in the Figure 12) was conditional upon certain tactical targets.

- **Melbourne-Sydney:** This route had the wild fluctuating demand pattern so could afford to have a lower fulfilment percentage (70-75%) as compared with the others giving greater operational flexibility to cope with variations in demand.

- **Perth-Sydney:** This route had a high percentage required (95%) because a secure demand pattern existed which had to have the off-setting high costs incurred by operation.
- **Gold Coast-Sydney:** This route had a varying demand pattern evidenced by the different percentages allowed (80%-65%) to meet the seasonal demand which necessitated peak and off-peak resource utilisation.

Constraint 1: Carrying Capacity Requirement (The carrying capacity is no less than a certain ratio of the passenger volume)	Projected Ratio for Carrying Capacity of Each Route
July (Melbourne - Sydney)	80%
August (Melbourne - Sydney)	80%
September (Melbourne - Sydney)	75%
October (Melbourne - Sydney)	75%
November (Melbourne - Sydney)	70%
December (Melbourne - Sydney)	70%
July (Gold Coast - Sydney)	80%
August (Gold Coast - Sydney)	80%
September (Gold Coast - Sydney)	75%
October (Gold Coast - Sydney)	75%
November (Gold Coast - Sydney)	70%
December (Gold Coast - Sydney)	65%
July (Perth - Sydney)	95%
August (Perth - Sydney)	95%
September (Perth - Sydney)	95%
October (Perth - Sydney)	95%
November (Perth - Sydney)	95%
December (Perth - Sydney)	95%

Figure 12: Constraint 1-Carrying Capacity Requirement

(The carrying capacity is no less than a certain ratio of the passenger volume)

2. Constraint 2 - Operational Stability: The number of flights scheduled per month must not exceed the average number of flights operated in the same month over the past two years (2023 & 2024), ensuring schedules are grounded in operational reality.

Keeping in mind the above constraints, basic assumptions were also made when constructing the model:

1. The seat capacity of the recommended aircraft (A330-300) is fixed at 297 seats per flight.
2. The available operational days and aircraft are sufficient to support the optimized schedule and would not be impacted by unanticipated events.
3. Since the Qantas annual report does not distinguish between fixed and variable costs, we use "Total Underlying Expenditure" as the basis for estimation. The operating cost

per flight will be estimated based on the Total Underlying Expenditure in FY2025 and the Total Number of Domestic Routes Operated by Qantas, resulting in an operational cost per route. The formula is as follows:

$$\text{Route Monthly Operational Cost} = \left(\frac{\text{total underlying expenditure FY2025}}{\text{number of domestic routes}} \right) \times \frac{1}{12} \times \text{projected ratio for flight operational cost}$$

4. Based on Qantas Annual report, flight operational cost is assumed to be 30% of the total underlying expenditure.
5. The Gold Coast-Sydney route is being used as the assumed route that represents the average for operational cost allocation because it has moderate passenger numbers and operational characteristics that represent the typical domestic route profile for Qantas.
6. The fare on the 1st of each month is taken as the representative monthly average price for each route.

4.3.4 Decision Variables and Objective Function

The decision variables were the number of flights allocated to each of the three routes (i = Melbourne-Sydney, Gold Coast-Sydney, Perth-Sydney) for each month from July to December 2025 (j = 1 to 6).

Objective Function:

The objective function was to maximize the total operational profit, calculated as the sum of profits from all routes and all months. Profit for each route-month is the revenue minus total costs.

The formula used can be written as below:

- i = Route index (1=Melbourne-Sydney, 2=Gold Coast-Sydney, 3=Perth-Sydney)
- j = Month index (1=July, 2=August, ..., 6=December)
- x_{ij} = Number of flights for route i in month j (Decision Variable)
- Number of Passengers $_{ij}$: The number of passengers on each aircraft (for example,

the A330-300 can accommodate 297 passengers) for each route

- CR_{ij}= Cancellation rate for route c in month j
- Fare_{ij}= Ticket fare for route for route i in month j (AUD)
- OpCost_{ij} = Operational cost per flight for route for route i

Maximize total operational profit for July-December 2025:

$$\text{Maximize } Z = \sum_{i=1}^3 \sum_{j=1}^6 [\text{Number of Passengers}_{ij} \times (1 - \text{CR}_{ij}) \times \text{Fare}_{ij} - \text{OpCost}_{ij}]$$

Model Implementation and Results:

The model was implemented in Excel and solved using the OpenSolver add-in. Appendix II looks at the results from the prescriptive analysis.

Objective Function Value:

The model ended up with a maximum total operational profit of AUD 979,129,825.42 for the July-December 2025 period.

Decision Variable Results: Optimal Flight Allocation

From the Excel solution the optimal flight allocation is displayed in Table 7 below.

Route / Month	July	August	September	October	November	December
Melbourne-Sydney	844	866	845	860	840	702
Gold Coast-Sydney	97	92	90	93	103	104
Perth-Sydney	200	202	201	206	194	199

Table 7. Optimal Monthly Flight Allocation (July–December 2025)

a) Aircraft are scheduled to be used at or near their maximum authorised capacity on the Melbourne - Sydney section from July to November, which indicates the profit and density of

its use. The reduction of flights in December (702), is in accordance with the expected decline in the number of passengers carried in that month.

b) The program on the Gold Coast - Sydney section lends itself well to variations. The allotment is made approximately to coincide with the expected demand curve, and the state of the traffic leads to the largest number of flights being programmed in December (114) to catch the seasonal traffic peak.

c) The utilization of the flight quota on the Perth - Sydney section is stable and high, indicating a uniformity of demand and profits for the whole of the period.

d) The solution is a confirmation that the recommended program is both operationally expedient (within limits in the historical range of flights) and commercially satisfactory (maximising profits so as to comply with agreements as to level of service).

5. Discussion

5.1 Strengths of Analysis

The analysis possesses many strengths that fit in well with the credibility and scholarly usefulness of its results. First of all, look closely at the implementation part, with the three-phase analytical framework (descriptive, predictive and prescriptive), and will thus ensure methodological integrity. In the descriptive part k-means clustering was used for segmenting the three domestic routes of Qantas according to passenger volumes and cancellations, thus showing the patterns of core demand. Predictively the expected volumes and cancellations are predicted, using both time series model and multiple regression model - thus accurately assessing future trends. In the prescriptive phase Integer Linear Programming is used to convert those results of predicting into action for Qantas scheduling the flights so as to gain the most from profits, achieving all this within the limits of the real-world.

5.1.2 Leveraging Real Company and Industry's Data

All the numbers are not pulled out of thin air but come from good data like BITRE and the annual report of Qantas itself. Such that the use of specific forecasts of their routes assists the empirical robustness of the regression, whilst elevating the correctness of its outcomes. Thus, arriving at the standard of rigor of severe empirical.

5.1.3 Alignment to Practice

The study has practical implications. An overhaul of the Integer linear programming models included a number of operational considerations. In particular, it incorporates previous flight restrictions, cost ratios which correlate the suggested solutions with the operational realities of Qantas. Furthermore, the prediction of AU\$979 million profit for H2 2025 that can be anticipated indicates applicability for strategic decision making for day to day operations.

In conclusion, the combination of analytical techniques, rigorous data practices and business applicability that is the study, presents a study with valuable contributions to the operations of the airline and quantitative research, at the same time keeping in mind both the scholarly depth to it and its real world implications.

5.2 Limitations of Analysis

This report's analysis considered variables related to Qantas' domestic operations, but several limitations should be acknowledged that assumptions made during the implementation. The passenger volumes (demand) used in both descriptive and predictive analysis is depended on publicly available industry level data to approximate Qantas' specific volumes. While this approach enabled a consistent dataset with market trend, it may not fully reflect Qantas' specific variations in demand and operational difference of routes with other airways. Additionally, although two considered variables were rescaled before clustering, the significant differences in passenger volumes across different routes may still dominate the clustering results. This dominance might weaken the model's ability to capture internal variations across different months for every single route.

In the predictive analysis part, data from January 2023 to June 2025 were used in order to minimize the negative effect of COVID-19. Although it reduces the abnormal fluctuations caused by lockdowns and travel restrictions, the available data range is restricted at the same

time. Additionally, forecasted cancellation rate with exponential smoothing and moving average method might be inaccurate since they are exclusively relying on historical data. Other factors about macroeconomic environment or exogenous shocks can also influence the accuracy of the forecast.

In the prescriptive analysis part, due to the complexity of the aviation industry, some simplifying assumptions were made to ensure a feasible and explicable optimisation model. First, each route was assumed to use the same aircraft model (A330-300) to fulfill the demand and we assumed all assigned flights would operate at full capacity. In real world situation, this assumption might be idealized. Second, a projected ratio for flight operational cost is assumed to account for 30% of the total cost and take the Gold Coast to Sydney route as the average level. Third, the air fare of economy and business class was treated as the same which ignoring the cabin differences that affect the profitability of the route. Thus, the profit calculated by the ILP model cannot fully reflect the profit in real life.

Overall, the limitations of this report mainly from the lack of Qantas' specific data and the model's inability to comprehensively consider all relevant factors that affect the operation of airlines.

5.3 Recommendations

According to the descriptive, predictive, and prescriptive examination of Qantas' three main domestic routes (Melbourne-Sydney, Gold Coast-Sydney, and Perth-Sydney), the primary cause of high cancellation rates and growing operating expenses is the existing mismatch between flight supply and demand. Specifically, high-demand routes suffer from insufficient capacity, while medium-to-low demand routes experience wasted flights. Therefore, if experience-based scheduling is used continuously, it may result in fuel and human resource waste during off-peak hours and capacity constraints during peak seasons.

5.3.1 Dynamic Adjustment of Flight Schedules

It is recommended to adopt a demand-forecast-based dynamic scheduling mechanism. For example, flight frequencies on the Melbourne-Sydney route could be increased during the busiest months (July-November) and promptly reduced in December in response to anticipated

drops in demand. In addition, the Gold Coast–Sydney route may take advantage of its seasonal peak by expanding its flight schedule during holidays to attract tourists, while the Perth–Sydney route might keep its frequencies constant to guarantee on-time service and profits.

5.3.2 Strengthen Operational Resilience and Cost Control

According to prescriptive analysis, the ILP optimization model is expected to provide AUD 979 million in profit under present constraints, and additional improvement is mostly dependent on lowering fixed operational expenses. Therefore, to improve schedule flexibility and maintenance efficiency, Qantas could first investigate fleet standardization techniques, such as giving priority to the deployment of A330-300 aircraft on important domestic routes. A tiered cost approach that distinguishes between routes with high and low demand could also be used to balance network coverage and profitability. Moreover, incorporating a data-driven crew scheduling system is also essential. Qantas may drastically cut overtime expenses and employee turnover rates by better aligning workforce levels with operating cycles. In the past, excessive growth resulted in rising overtime costs and higher employee turnover, both of which negatively impacted service dependability and brand reputation.

6. Conclusion

In conclusion, the aviation industry is entering a period of recovery and re-competition after the pandemic, which makes flight optimisation more essential to Qantas' operational efficiency and profitability. By leveraging descriptive, predictive, and prescriptive analysis, this report offers reasonable suggestions for future flight allocation for Qantas. The analysis only focused on three representative domestic routes, which limits the feasibility of the research results on all Qantas routes. Nevertheless, the modeling process provides an extensible framework that enables Qantas to optimize flight allocation while considering the impact on profit and enhance its competitive position in the constantly developing aviation industry.

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Appendix 1: Predictive

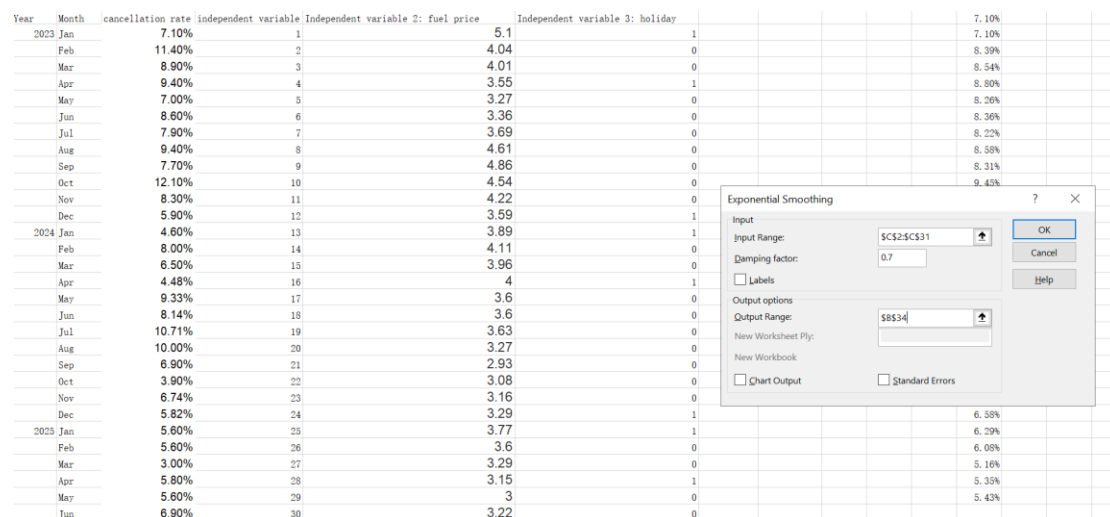


Figure1: Melbourne-Sydney Route Cancellation Rate Forecast (ES)

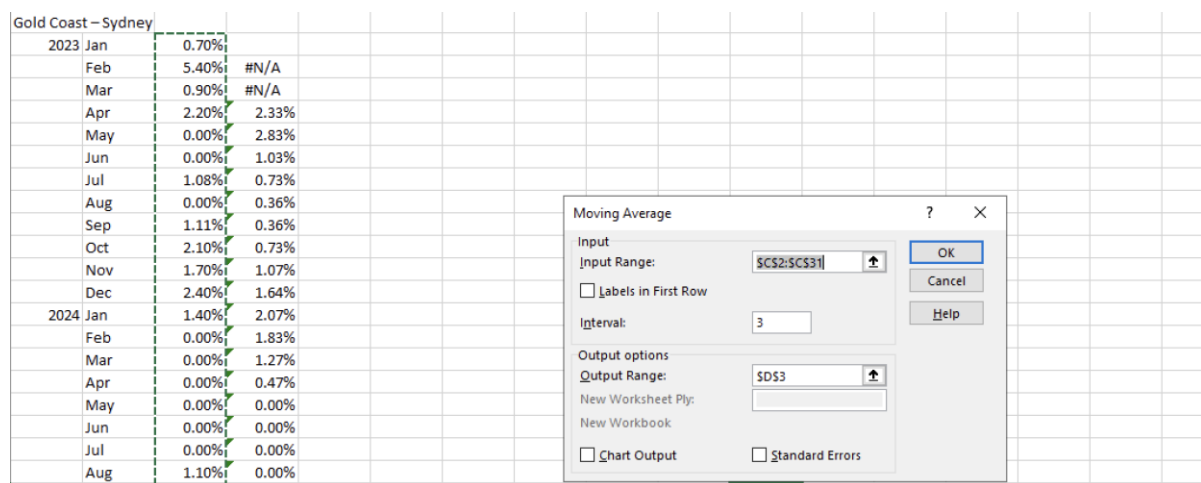


Figure 2: Gold Coast-Sydney Route Cancellation Rate Forecast (MA)

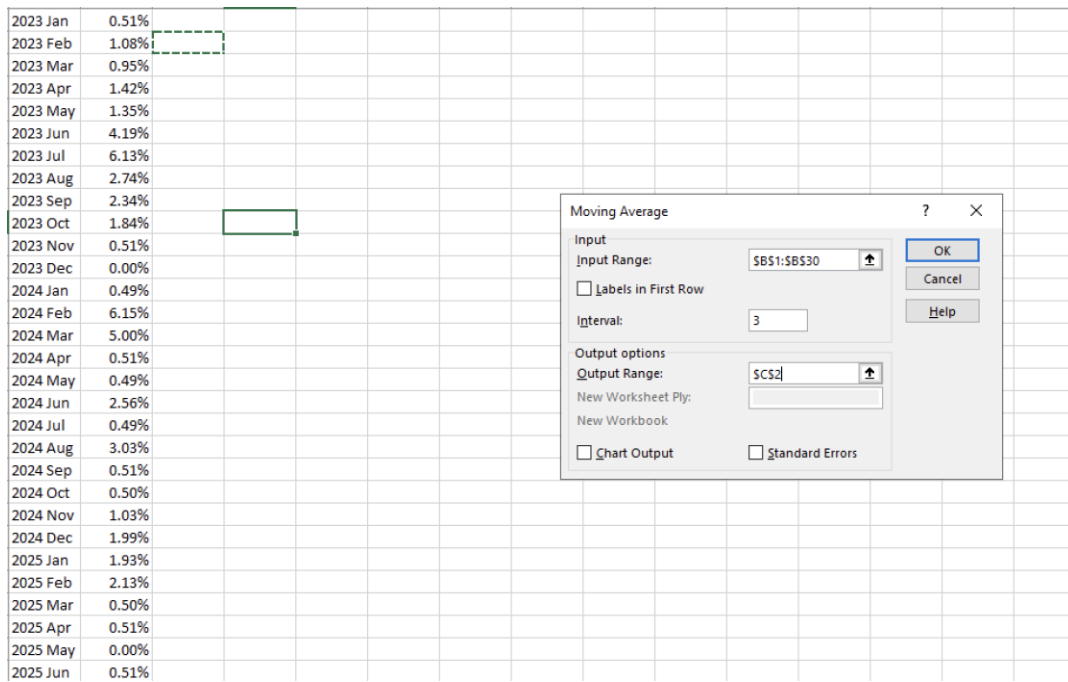


Figure 3. Perth-Sydney Route Cancellation Rate Forecast (MA)

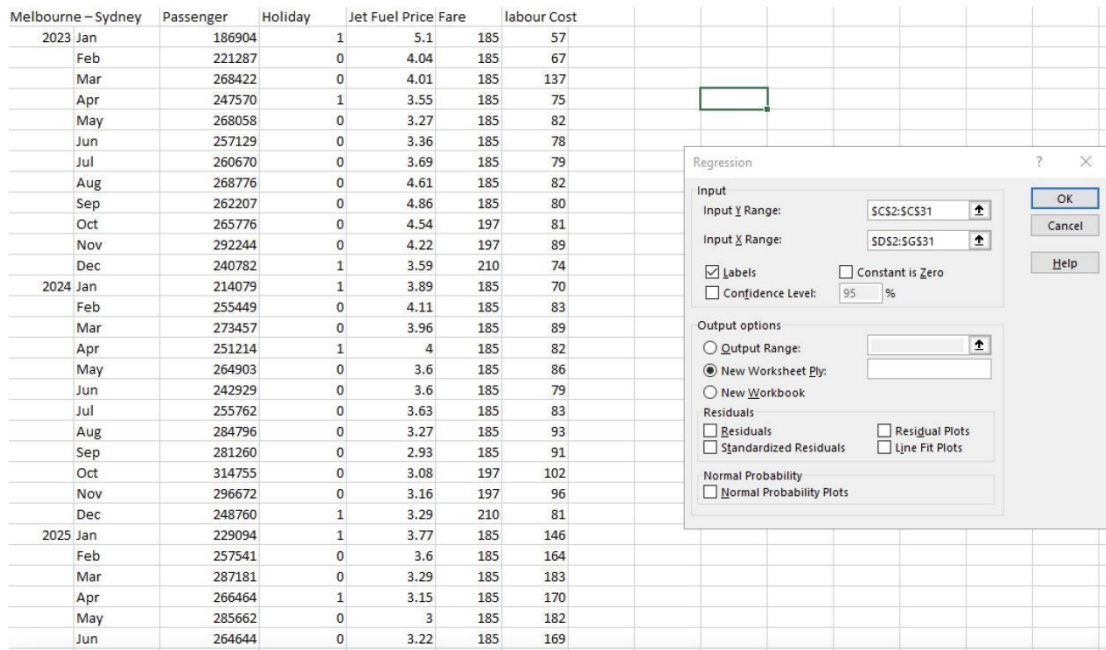


Figure 4. Melbourne-Sydney Route Passengers Carried Forecast (Multiple Regression)

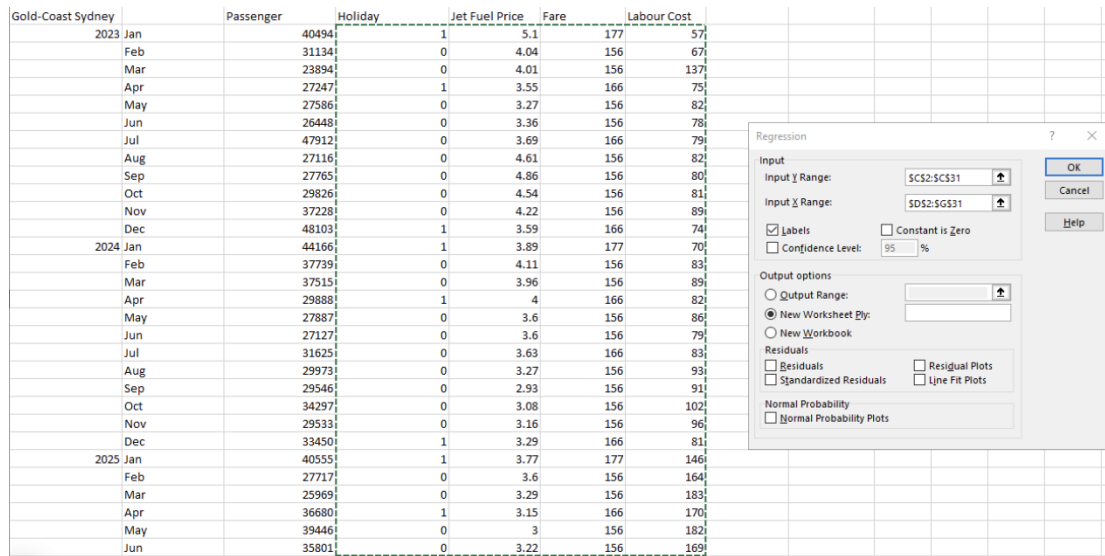


Figure 5. Gold Coast-Sydney Route Passengers Carried Forecast (Multiple Regression)

Appendix 2: Prescriptive: ILP Model's Core Variables

Elements	Explanation
Decision Variable	The number of flights on the three routes (Melbourne-Sydney, Gold Coast-Sydney, Perth - Sydney) from July to December 2025 (a total of $3 \times 6 = 18$ variables)
Objective Function	Maximizing the total profit for July-December 2025 = Σ (profit of a single route in a single month $\times$ number of flights in that month) - Monthly profit of single route = (Projected operational revenue of each route - estimated operational cost of each route)
Two Constraints	C1: Carrying Capacity $\geq$ certain ratio of the predicted demand (passenger volume) C2: The number of flights $\leq$ the average of the same period in the past 2 years

Appendix 3: ILP Model Solver's Implementation Results

Predicted Number of Passengers						
Route/Month	July	August	September	October	November	December
Melbourne - Sydney	280021	300063	291196	296341	326906	269550
Gold Coast - Sydney	32839	31474	29906	31986	39945	51604
Perth - Sydney	48238	48719	51614	52597	54383	54322
Flight Ticket Fare (\$)						
Route/Month	July	August	September	October	November	December
Melbourne - Sydney	185	185	185	197	197	210
Gold Coast - Sydney	166	156	156	156	156	166
Perth - Sydney	351	351	429	372	351	435
Cancellation Rate						
Route/Month	July	August	September	October	November	December
Melbourne - Sydney	5.87%	5.87%	7.37%	5.87%	5.87%	8.87%
Gold Coast - Sydney	0.53%	0.18%	0.24%	0.32%	0.25%	0.27%
Perth - Sydney	0.34%	0.28%	0.38%	0.33%	0.33%	0.33%
Seat Numbers for the Recommended Aircraft (A330-300)						
Route/Seat Type	Business Class	Economy Class	Total			
Melbourne - Sydney	28	269	297			
Gold Coast - Sydney	28	269	297			
Perth - Sydney	28	269	297			
Number of Flights Flown (2023)						
Route/Month	July	August	September	October	November	December
Melbourne - Sydney	846	878	826	808	863	708
Gold Coast - Sydney	92	93	89	95	117	123
Perth - Sydney	199	213	209	213	197	201
Number of Flights Flown (2024)						
Route/Month	July	August	September	October	November	December
Melbourne - Sydney	842	855	864	912	817	696
Gold Coast - Sydney	103	91	91	102	89	105
Perth - Sydney	202	192	194	200	192	197
Number of Flights Flown (Average)						
Route/Month	July	August	September	October	November	December
Melbourne - Sydney	844	866.5	845	860	840	702
Gold Coast - Sydney	97.5	92	90	98.5	103	114
Perth - Sydney	200.5	202.5	201.5	206.5	194.5	199
Projected Operational Revenue (\$)						
Route/Month	July	August	September	October	November	December
Melbourne - Sydney	48762996.95	52253120.85	49900948.14	54952319.31	60620173.71	51584582.15
Gold Coast - Sydney	5422382.25	4901106.10	4654139.19	4973848.59	6215841.45	8543135.09
Perth - Sydney	16873970.77	17052487.97	22058264.86	19501515.92	19025441.17	23552090.77
Important Factors for Consideration						
Total Underlying Expenditure for Qantas Domestic in FY2025 (\$)	21184000000					
Number of Months per Year	12					
Number of Domestic Routes for Qantas (Including Jetstar)	98					
Projected Ratio for Flight Operational Cost	30%					
Assumed Route Representing the Average	Gold Coast - Sydney					
Estimated Operational Cost (\$)						
Route/Month	July	August	September	October	November	December
Melbourne - Sydney	48598421.16	52076765.85	49732531.75	54766854.47	60415579.78	51410493.44
Gold Coast - Sydney	5404081.63	4884564.80	4638431.41	4957061.78	6194862.90	8514301.89
Perth - Sydney	16817020.89	16994935.58	21983817.91	19435698.04	18961230.04	23472602.14
Projected Operational Profit (\$)						
Route/Month	July	August	September	October	November	December
Melbourne - Sydney	164575.79	176355.01	168416.39	185464.84	204593.93	174096.71
Gold Coast - Sydney	18300.62	16541.30	15707.78	16786.81	20978.55	28833.20
Perth - Sydney	56949.88	57552.38	74448.95	65817.89	64211.13	79488.63
Decision Variable (Number of Flights)						
Route/Month	July	August	September	October	November	December
Melbourne - Sydney	844	866	845	860	840	702
Gold Coast - Sydney	97	92	90	98	103	114
Perth - Sydney	200	202	201	206	194	199
Objective Function (Maximum Total Operational Profit) (\$)						
			979129825.42			

Constraint 1: Carrying Capacity Requirement (The carrying capacity is no less than a certain ratio of the passenger volume)							
	Left hand Side (LHS)		Right hand Side (RHS)				Projected Ratio for Carrying Capacity of Each Route
July (Melbourne - Sydney)	235953.7884	>=	224016.80				80%
August (Melbourne - Sydney)	242104.2425	>=	240050.40				80%
September (Melbourne - Sydney)	232468.6795	>=	216397.00				75%
October (Melbourne - Sydney)	240426.846	>=	222255.75				75%
November (Melbourne - Sydney)	234835.324	>=	228834.20				70%
December (Melbourne - Sydney)	190000.5652	>=	188865.00				70%
July (Gold Coast - Sydney)	28956.3123	>=	26271.20				80%
August (Gold Coast - Sydney)	27274.8188	>=	25175.20				80%
September (Gold Coast - Sydney)	26955.848	>=	22426.50				75%
October (Gold Coast - Sydney)	29012.8698	>=	23889.50				75%
November (Gold Coast - Sydney)	30514.5225	>=	27961.50				70%
December (Gold Coast - Sydney)	33798.3634	>=	32042.60				65%
July (Perth - Sydney)	58186.04	>=	45026.10				80%
August (Perth - Sydney)	58626.0185	>=	46263.05				80%
September (Perth - Sydney)	58470.1514	>=	49033.30				90%
October (Perth - Sydney)	60980.0594	>=	49967.15				90%
November (Perth - Sydney)	57427.8606	>=	51663.65				90%
December (Perth - Sydney)	58907.9601	>=	51605.90				95%
Constraint 2: Number of Flights Restriction (The number of flights is less or equal to the average of the past 2 years)							
	Left hand Side (LHS)		Right hand Side (RHS)				
July (Melbourne - Sydney)	844	<=	844				
August (Melbourne - Sydney)	866	<=	866.5				
September (Melbourne - Sydney)	845	<=	845				
October (Melbourne - Sydney)	860	<=	860				
November (Melbourne - Sydney)	840	<=	840				
December (Melbourne - Sydney)	702	<=	702				
July (Gold Coast - Sydney)	97	<=	97.5				
August (Gold Coast - Sydney)	92	<=	92				
September (Gold Coast - Sydney)	90	<=	90				
October (Gold Coast - Sydney)	98	<=	98.5				
November (Gold Coast - Sydney)	103	<=	103				
December (Gold Coast - Sydney)	114	<=	114				
July (Perth - Sydney)	200	<=	200.5				
August (Perth - Sydney)	202	<=	202.5				
September (Perth - Sydney)	201	<=	201.5				
October (Perth - Sydney)	206	<=	206.5				
November (Perth - Sydney)	194	<=	194.5				
December (Perth - Sydney)	199	<=	199				

Objective Cell (Max)

Cell	Name	Original Value	Final Value
\$D\$79	Objective Function (Maximum Total Operational Profit) (\$) July	0	979129825.4

Variable Cells

Cell	Name	Original Value	Final Value	Integer
\$D\$73	Melbourne - Sydney July	0	844	Integer
\$E\$73	Melbourne - Sydney August	0	866	Integer
\$F\$73	Melbourne - Sydney September	0	845	Integer
\$G\$73	Melbourne - Sydney October	0	860	Integer
\$H\$73	Melbourne - Sydney November	0	840	Integer
\$I\$73	Melbourne - Sydney December	0	702	Integer
\$D\$74	Gold Coast - Sydney July	0	97	Integer
\$E\$74	Gold Coast - Sydney August	0	92	Integer
\$F\$74	Gold Coast - Sydney September	0	90	Integer
\$G\$74	Gold Coast - Sydney October	0	98	Integer
\$H\$74	Gold Coast - Sydney November	0	103	Integer
\$I\$74	Gold Coast - Sydney December	0	114	Integer
\$D\$75	Perth - Sydney July	0	200	Integer
\$E\$75	Perth - Sydney August	0	202	Integer
\$F\$75	Perth - Sydney September	0	201	Integer
\$G\$75	Perth - Sydney October	0	206	Integer
\$H\$75	Perth - Sydney November	0	194	Integer
\$I\$75	Perth - Sydney December	0	199	Integer

Appendix 4: Literature Review Summary

Article 1	Lonzius, C., & Lange, A. (2024). Aircraft routing clusters and their impact on airline delays. <i>Journal of Air Transport Management</i> , 114. https://doi.org/10.1016/j.jairtraman.2023.102493
Synopsis	In this article, a clustering method based on the number of airports served by each flight route, the number of flights each flight route has and the number of non-shuttle flights each flight route has been proposed, which caused 29 clusters to be formed which are ranged from routes that have only two airports served and progressive up to routes that are air transport routes which are multiairport hub and spoke. The main result is that the routes classified as shuttle routes have the greatest risk of delay and the hub and spoke routes have the least risk of delay.
Strengths	With the clustering model methodology presented in this paper, we can classify used by Qantas and thus identify shared patterns and distinctiveness in their operations. Also, classification results obtained through clustering can help us learn about performance under varying routes and operating conditions in a more nuanced sense and provide more targeted analytical implications for study.
Weaknesses	For Qantas routes, reliable clustering is difficult to achieve due to insufficient data, requiring manual intervention. Furthermore, the analysis lacks consideration for extreme scenarios, such as sudden passenger demand drops due to events like pandemics.
Summary	The three routes grouping model presented in this paper provides Qantas with an analytical structure that relates to delay and cost. Determining the type of route clusters, it helps manage delay cost control explicitly for the purpose of reducing operating costs. However, to maximize the benefits of this data-driven decision-making model, it should be complemented with Qantas' past operating cost data and real-time passenger flow adjustments.

Article 2	Yu, D., Dong, S., & Yao, S. (2022). Improvement of K-Means algorithm and its application in air passenger grouping. <i>Computational Intelligence and Neuroscience</i> , 2022, 1–13. https://doi.org/10.1155/2022/3958423
Synopsis	This article shows K-means is trying to place n observations into k groups such that every observation will fall into the cluster with the nearest mean (cluster centers are also known as centroids). It will choose k random starting centroids, then continually assign data points to the nearest centroid and update the centroids until it stabilizes.
Strengths	The K-means model boasts notable strengths in three key aspects. In terms of simplicity, it is easy to understand and implement, which makes it highly accessible for beginners engaging in data analysis. Regarding efficiency, it can handle large datasets relatively well, and this performance is particularly prominent when the number of clusters k is small. When it comes to interpretability, the resulting clusters from K-means are straightforward to interpret, as each centroid clearly represents the center of a corresponding cluster.
Weaknesses	Random initial picks can result in different outcomes of clustering. A bad initial pick can cause the algorithm to converge to a local minimum rather than the global maximum. The algorithm lacks any built-in mechanism for deciding the optimal k, and an incorrect k value can result in over-clustering or under-clustering.
Summary	K-means is a popular and simple clustering model. It is efficient, interpretable and easy to use. However, it has flaws in the shape of sensitivity to initial points, difficulty in determining the number of clusters, and vulnerability to outliers. Despite its weaknesses, it is a standard exploratory data analysis and clustering technique whenever there is roughly spherical and well-separated structure in the clusters.

Article 3	Lee, S., Hong, Y., & Kim, Y. (2019). Optimal scheduling algorithm in point merge system including holding pattern based on mixed-integer linear programming. <i>Proceedings of the Institution of Mechanical Engineers Part G Journal of Aerospace Engineering</i>, 234(10), 1638–1647. https://doi.org/10.1177/0954410019830172
Synopsis	This article proposes a Point Merrier System (PMS) scheduling algorithm with a "wait-return module." This model reduces air delays and minimizes air traffic controller intervention by imposing virtual constraints and corrections.
Strengths	The study presents a vigorous optimization approach supported by Monte Carlo simulations. It integrates holding patterns into PMS and presents a more realistic solution to congestion, as well as improving efficiency and controller workload management.
Weaknesses	The model is computationally intensive, with performance degrading as problem size. Its 30-minute scenario focus and simulation-based nature limit use to real-world, long-term operations. Fairness between aircraft in delay assignment is addressed only indirectly.
Summary	This contribution provides an additional decision-support system for air traffic control through the integration of holding maneuvers into PMS algorithms. With a great promise in reducing delays and controller workload, there is still room for improvement in scalability, uncertainty consideration, and fairness in scheduling.

Article 4	Abdelghany, A., Abdelghany, K., Guzhva, V. S., & Kai, M. (2025). Unraveling endogeneity in seat capacity and fares: Time series econometric models for airline origin-destination passengers forecasting. <i>Journal of Air Transport Management</i>. https://doi.org/10.1016/j.jairtraman.2025.102832
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Synopsis	The article focuses on applying Time-series models to forecast airline origin-destination (O-D) passenger numbers. It develops three key time-series models: the baseline Seasonal Autoregressive Integrated Moving Average (SARIMA), Vector Autoregression (VAR) with endogenous variables, and SARIMAX with exogenous variables, and tests them on O-D pairs like LAX-JFK and IST-LHR, as well as a large sample of 2000 O-D pairs.
Strengths	The time-series models in this article really shine when it comes to capturing the temporal patterns that matter so much for airline forecasting. SARIMA is great at handling those seasonal ups and downs in passenger numbers. Then there's SARIMAX-by bringing in exogenous variables like seat capacity, it's way more accurate at forecasting than VAR. That makes it a better fit for real-world airline operations.
Weaknesses	The VAR model, which is one of the time-series based forecasting models in the literature, is quite poor in forecasting when seat capacity and fares are considered as endogenous variables. It has higher SMAE, MAPE and RMSE than the baseline SARIMA, which indicates that it probably introduces unnecessary noise into passenger forecasting. The SARIMAX model also has some problems. If we use both seat capacity and fares as exogenous variables at the same time we may run into multicollinearity in the structure or overfitting in the evaluations. This can cause greater forecasting errors.
Summary	In this paper we find that time-series based forecasting is useful for airline O-D passenger forecasting. There is a distinct difference in result output that varies depending on how the prospective variables are defined and managed. SARIMAX at least in the use of using the seat capacity as an exogenous variable is more successful than either the VAR model or even the baseline SARIMA forecasting technique in comparing accuracy of forecasting and how productive economically it is to use the techniques at all. This is henceforth suggested that the importance of picking the right variables to forecast either O-D passenger output is of vital importance along with the systems to forecast.

Article 5	Zhou, Z., Liang, C., Chou, M. C., Chou, W., & Yu, G. (2020). Airline planning and scheduling: Models and solution methodologies. <i>Frontiers of Engineering Management</i> , 7(1), 1–21. https://doi.org/10.1007/s42524-020-0093-5
Synopsis	The paper considers the basic models used in airline planning which involves the manner in which airlines provide their fleets, route their aircraft and the way in which they schedule their crews. It looks at the optimisation methods such as integer linear programming, network flow, decomposition and heuristics and also looks at robust and stochastic methods dealing with random demand and operational disruptions. The authors stress the point that it is important that the various levels of planning be designed in such a way as to achieve a minimum of cost and a reliability of operations.
Strengths	A major advantage of this paper is the breadth of its coverage of the models applied to airline scheduling. It gives managers a good overall view of the various methods applicable to them. It stresses the interdependence of fleet planning, routing and crew planning which shows that integrated methods are more efficient than optimisation of any one of these areas separately. Since the robustness and flexibility of these models is stressed, it can be seen that the models are either more likely to be applicable in the case of the airline environment of uncertainty or at least to airline management. The fact that such things as maintenance requirements and airport bottlenecks are mentioned tends to emphasise the devoteness of this research in the real world.
Weaknesses	The paper is however somewhat theoretical in nature with the result that it is valid in little if any in the nature of concrete clarification as to its applicability to the individual problem of the various particular airlines such as Qantas. Many of the models assume approximately constant demand or static schedules which does not really reflect the messy real world complexity of airline operation. Little is said in the way of implementation difficulties such as whether the required weather data will be able or whether the models will be susceptible to scale. There is no discussion of the way the various airlines could put them into

	operation. Although the environment and sustainability items are assuming increasing importance, these items are only superficially touched on.
Summary	In fact, this paper is really a good basic foundation on the subject of airline planning and scheduling, explaining the way integrated and robust solutions can increase fleet assignment. The one major defect is in the line of real case applications, and for that matter the use that they also make of the sustainability features. However it does constitute a good basis for decision-makers to work from in their management of airlines. For Qantas the use of the models to which reference has been made would lead to recommended efficiency, increased elasticity of operations and application to long-term strategic planning.

Article 6	Ahmed, M. B., Mansour, F. Z., & Haouari, M. (2018). Robust integrated maintenance aircraft routing and crew pairing. <i>Journal of Air Transport Management</i> , 73, 15-31.
Synopsis	This study presents a MILP model which integrates maintenance routing of aircraft with crew pairing opportunities. It builds in uncertainties such as variable maintenance times and crew availability with the objective of improving operational efficiency and robustness. For Qantas, it directly deals with such issues as flight cancellations, MILP related issues and labour shortages.
Strengths	It integrates maintenance and crew planning resulting in cost reductions and operational disruption risk reductions. The MILP Module provides workable schedules even in uncertain situations providing proactive support in areas of capacity and labour planning for Qantas.
Weaknesses	The model is based on a number of simplified assumptions and involves intensive computations which will limit its application to Qantas' large network. It also does not take account of passenger experience or industrial relations.

Summary	By using robust integrated MILP in this way this research demonstrates how it is possible for airlines to consider simultaneously the planning of aircraft maintenance activities with crew pairings, in conjunction with a building of their robustness with respect to uncertainty. For Qantas the use of such methods might lead to reductions in cancellations and delays caused by labour shortages and these numerical methods in turn, might alleviate the problems caused by maintenance disruptions, leading to enhanced OTP. However, full practical implementation would require a further adaptation of the model so as to deal with its complexity, labour agreements and passenger performance metric focused considerations.
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Appendix 5: Group Meeting

Meeting1	Saturday, 6th September 2025
Agenda	Review of Assignment 2 requirements and marking criteria. Discussion of potential industries and companies. Final decision on selected company and business problems. Confirmation of tasks, role allocation, and next meeting schedule. Discussion Notes Assignment requirements reviewed: 5,000 words, group report due Week 11, presentation in Week 12. Must include three types of analytics: Descriptive, Predictive, Prescriptive. Minimum six peer-reviewed journal articles for Literature Review. Appendices must include meeting notes. Intro + Literature Review: Yufei & Minyue Collect Data: Yuhong, Junlin & Jiayi Next Meeting Date: Sunday, 14th September 2025
Absent	Yuhong Xu

Meeting2	Sunday, 15th September 2025
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Agenda	Discussion on improvement of Literature Review New Research of peer-reviewed papers for Literature Review Areas of Data collection Next meeting schedule. Discussion Notes : Literature Review: Pay attention to the applications of analytics methods in the peer-reviewed papers; what we can get from the papers which can give us insights of how to solve problems by analyzing certain factors. Data Collection: We will mainly focus on Top 20 domestic routes Passengers Carried (demand) Source: Bureau of Infrastructure and Transport Research Economics (BITRE). Accessibility: Publicly available on BITRE’s aviation statistics database. Limitation: Only industry-wide data is provided (per route, per month). Qantas-specific passenger numbers are not available. Approach: To approximate Qantas’ passengers carried, we calculate proportion of Qantas flights = (Qantas’ actual flights flown ÷ total industry flights flown). Then, apply this ratio to industry-wide passenger numbers to get approximately Qantas’ passengers carried. (Many assumptions are needed) - Qantas' Sched. & actual flights flown, on-time performance, cancellation rates Source: BITRE Accessibility: Publicly available; contains airline-specific data of Qantas. Data that might be needed: - Fuel price Source: U.S. Energy Information Administration (EIA) – U.S. Gulf Coast Kerosene-Type Jet Fuel Spot Price Accessibility: Publicly available Limitation: This is a U.S. benchmark. Australian carriers may pay 5–10% premium due to freight costs and regional market conditions. - Labour expense Source: Qantas Annual Reports (2023–2025) Note: Annual reports disclose consolidated Group-level “Employee benefits expense”, not Airways-only. So se Group-level figures as proxy - Maintenance Expense Source: Qantas Annual Reports (2023–2025).
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	Note: Similar to labour, only Group-level “Repairs and maintenance” available. Method: Apply proportional allocation across months using Qantas Airways flights operated (BITRE) as weight. - Extreme Weather Days (Dummy) Source: Bureau of Meteorology (BOM) monthly climate summaries Note: Record extreme rainfall, heat, storm/flood days. - Air Traffic Control Delay (Dummy) Source: Airservices Australia – Monthly Network Performance Reports. - Industrial Action (Dummy) Source: Qantas Newsroom, union announcements, and major media outlets (ABC, Guardian, SMH). Method: Confirm industrial action events (strikes, stoppages). Dummy = 1 if event occurred in that month Next Meeting Sunday, 21st September 2025
Attendees	All Members

Meeting3	Saturday, 20th September 2025
Agenda	key Discussion Points 1.1 Research Problem Adjustment: the team acknowledged that the initial scope was too broad and complex and agreed to narrow the focus to aircraft type allocation on Qantas’ top 10 domestic routes, with optimization objectives set around operating costs. 1.2 Literature review: Find more studies about Clustering, Time series, and Regression model 1.3 Analytical Framework - Descriptive analysis: K-means clustering - Predictive analysis: (possible choice) apply time series and regression to forecast demand and route performance. - Prescriptive analysis: Develop an optimization model using MILP Work allocation - Descriptive: Jiayi --ddl: Sep 29 - Predictive: Yufei, Yuhong -- ddl: Sep 29 - Prescriptive: Junlin, Minyue -- ddl: Oct 5

	Next Meeting Saturday 27th September 2025
Attendees	All Members

Meeting4	Saturday, 27th September 2025
Agenda	Based on the data, ten routes were too numerous to analyze. Therefore, we decided to consider only three representative routes. Discuss the models used in the three analyses and their specific implementation: Descriptive: K-means Clustering Predictive: Time-series Model + Multiple Regression Prescriptive: ILP Our team members also made a final research outline for each analysis section to ensure the report's logical structure and coherence. After data cleaning and verification, the model and analysis steps will be further strengthened. Next Meeting Friday 12th October 2025
Attendees	All Members

Meeting5	Friday, 12th October 2025
Agenda	Comprehensively discuss and research the completed work and make detailed revisions and improvements. Discuss the presentation format Record key Excel calculations Organize and consolidate Excel data Modify and check the final report
Attendees	All Members